

Falcon Post-Quantum Signature Scheme

Function-Level Profiling Report
-00 vs -03 · 1000 iterations · gprof

Falcon-512 / Falcon-1024

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1 Overview

This report presents a function-level profiling analysis of the **Falcon** post-quantum digital signature scheme (NIST PQC Round 3 finalist, lattice-based). Two parameter sets are compared: **Falcon-512** (NIST Level 1) and **Falcon-1024** (NIST Level 5). Each operation was executed **1000** times and profiled with **gprof** under two compiler settings: **-O0 -fno-inline -pg** and **-O3 -fno-inline -pg**.

Functions are grouped into three categories:

- **Hash / PRNG** — SHAKE-256 (Keccak), PRNG refill, hash-to-point.
- **Integer Arithmetic** — Modular NTT (`modp_*`, `mq_*`), multi-precision integers (`zint_*`), Gaussian sampler (`mkgauss`).
- **Floating-Point / FFT** — Custom 53-bit FPR type (`fpr_*`, `FPR`), FFT/iFFT, polynomial operations.

2 Runtime Summary: O0 vs O3

Table 1: Total cumulative runtime (seconds, 1000 iterations)

Variant	Operation	O0 (s)	O3 (s)	Speedup
Falcon-512	KeyGen	51.96	11.74	4.43 ×
Falcon-512	Sign	15.93	3.24	4.92 ×
Falcon-512	Verify	0.19	0.05	3.80 ×
Falcon-1024	KeyGen	143.26	34.31	4.18 ×
Falcon-1024	Sign	34.24	6.81	5.03 ×
Falcon-1024	Verify	0.59	0.08	7.38 ×

3 Category Weight per Operation

The following tables show what fraction of total profiled time is spent in each functional category for O0 and O3 builds.

3.1 Category Breakdown — -O0

Table 2: Category time share -O0 (%)

Variant	Operation	Hash	Integer	FP/FFT	Other
Falcon-512	KeyGen	5.8%	31.9%	61.1%	1.2%
Falcon-512	Sign	1.4%	2.1%	96.2%	0.3%
Falcon-512	Verify	0.0%	78.9%	21.1%	0.0%
Falcon-1024	KeyGen	3.2%	33.3%	62.5%	1.0%
Falcon-1024	Sign	1.8%	2.0%	96.0%	0.3%
Falcon-1024	Verify	13.6%	54.2%	28.8%	3.4%

3.2 Category Breakdown — -O3

Table 3: Category time share -O3 (%)

Variant	Operation	Hash	Integer	FP/FFT	Other
Falcon-512	KeyGen	8.3%	39.0%	50.9%	1.8%
Falcon-512	Sign	2.8%	2.8%	94.4%	0.0%
Falcon-512	Verify	40.0%	60.0%	0.0%	0.0%
Falcon-1024	KeyGen	4.5%	40.6%	53.2%	1.6%
Falcon-1024	Sign	1.2%	3.2%	95.6%	0.0%
Falcon-1024	Verify	25.0%	62.5%	12.5%	0.0%

4 Per-Operation Function Detail

Each subsection shows the top functions per category for both O0 and O3 builds. Column headers: % = percentage of total profiled time, **self(s)** = self seconds, **calls** = call count.

4.1 Falcon-512 — KeyGen

-O0 build (total: 51.960 s)

Hash / PRNG — 5.8% of total

Function	%	self(s)	calls
KeccakF1600_StatePermute	3.75	1.9500	2.7M
keccak_inc_squeeze	1.51	0.7900	46.1M
get_rng_u64	0.35	0.1800	46.1M
shake256_inc_squeeze	0.15	0.0800	46.1M
PQCLEAN_randombytes	0.00	0.0000	1.0K
keccak_inc_absorb	0.00	0.0000	1.0K
keccak_inc_finalize	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	1.0K
shake256_inc_absorb	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 31.9% of total

Function	%	self(s)	calls
modp_montymul	11.90	6.1800	980.6M
modp_add	4.89	2.5400	686.8M
mkgauss	2.54	1.3200	11.5M
zint_mod_small_unsigned	2.25	1.1700	50.0M
zint_add_scaled_mul_small	1.92	1.0000	18.1M
modp NTT2_ext	1.44	0.7500	2.0M

Function	%	self(s)	calls
modp_sub	1.19	0.6200	368.2M
modp_R2	0.89	0.4600	3.8M
zint_add_mul_small	0.69	0.3600	21.7M
zint_finish_mod	0.48	0.2500	895.8K
zint_norm_zero	0.46	0.2400	21.1M
modp_iNTT2_ext	0.46	0.2400	1.8M
mq_montymul	0.35	0.1800	18.3M
zint_bezout	0.35	0.1800	1.0K
zint_mul_small	0.34	0.1700	1.5M

Floating-Point / FFT — 61.1% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_fpr_add	30.78	15.9900	301.7M
FPR	11.01	5.7200	662.7M
PQCLEAN_FALCON512_CLEAN_fpr_mul	10.61	5.5100	305.8M
PQCLEAN_FALCON512_CLEAN_fpr_scaled	3.45	1.7900	53.3M
fpr_ulsh	2.06	1.0700	301.7M
fpr_ursh	1.35	0.7000	301.7M
PQCLEAN_FALCON512_CLEAN_FFT	0.52	0.2700	609.0K
fpr_sub	0.38	0.2000	121.0M
PQCLEAN_FALCON512_CLEAN_fpr_div	0.35	0.1800	2.0M
PQCLEAN_FALCON512_CLEAN_iFFT	0.25	0.1300	303.6K
fpr_rint	0.16	0.0900	4.3M
fpr_of	0.07	0.0400	53.3M
fpr_neg	0.03	0.0100	8.1M
fpr_sqr	0.02	0.0100	6.8M
PQCLEAN_FALCON512_CLEAN_poly_mul_fft	0.02	0.0100	578.6K

Other — 1.2% of total

Function	%	self(s)	calls
modp_div	0.25	0.1300	1.0M
poly_big_to_fp	0.25	0.1300	594.7K
modp_ninv31	0.10	0.0500	2.8M
modp_mkgm2	0.10	0.0500	1.0M
poly_sub_scaled	0.10	0.0500	520.6K
solve_NTRU_intermediate	0.08	0.0400	7.0K
fpr_lt	0.07	0.0400	6.5M
poly_sub_scaled_ntt	0.06	0.0300	56.1K
make_fg_step	0.06	0.0300	44.3K
mq_div_12289	0.04	0.0200	530.7K
make_fg	0.04	0.0200	8.0K
modp_R	0.02	0.0100	6.8M
PQCLEAN_FALCON512_CLEAN_poly_adj_fft	0.02	0.0100	25.3K
PQCLEAN_FALCON512_CLEAN_poly_invnorm2_fft	0.02	0.0100	13.7K
PQCLEAN_FALCON512_CLEAN_poly_add_mul_adj_fft	0.02	0.0100	1.0K

-03 build (total: 11.740 s)

Hash / PRNG — 8.3% of total

Function	%	self(s)	calls
KeccakF1600_StatePermute	4.77	0.5600	2.6M
keccak_inc_finalize	1.70	0.2000	43.8M
get_rng_u64	1.70	0.2000	43.8M
shake256_inc_squeeze	0.09	0.0100	43.8M
PQCLEAN_randombytes	0.00	0.0000	1.0K
keccak_inc_absorb	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	1.0K
shake256_inc_absorb	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 39.0% of total

Function	%	self(s)	calls
modp_montymul	18.48	2.1700	968.1M
mkgauss	4.00	0.4700	10.9M
modp_add	2.81	0.3300	686.0M
modp_sub	1.96	0.2300	366.5M
zint_add_scaled_mul_small	1.87	0.2200	18.1M
zint_rebuild_CRT_constprop.2	1.53	0.1800	36.2M
zint_finish_mod	1.11	0.1300	895.8K
zint_mod_small_unsigned	1.02	0.1200	49.5M
zint_co_reduce_mod	1.02	0.1200	447.9K
zint_norm_zero	0.85	0.1000	20.1M
modp_iNTT2_ext	0.85	0.1000	1.8M
zint_add_mul_small	0.68	0.0800	21.7M
mq_montymul	0.51	0.0600	17.8M
poly_big_to_small	0.43	0.0500	20.1M
modp NTT2_ext	0.43	0.0500	220.0K

Floating-Point / FFT — 50.9% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_fpr_add	27.68	3.2500	294.8M
PQCLEAN_FALCON512_CLEAN_fpr_mul	11.93	1.4000	300.1M
PQCLEAN_FALCON512_CLEAN_fpr_scaled	3.15	0.3700	53.0M
FPR	3.07	0.3600	649.8M
fpr_trunc	1.70	0.2000	294.8M
fpr_ursh	1.45	0.1700	294.8M
PQCLEAN_FALCON512_CLEAN_fpr_div	0.85	0.1000	1.9M
PQCLEAN_FALCON512_CLEAN_iFFT	0.34	0.0400	303.3K

Function	%	self(s)	calls
fpr_rint	0.26	0.0300	4.3M
PQCLEAN_FALCON512_CLEAN_FFT	0.26	0.0300	609.0K
fpr_sub	0.17	0.0200	117.7M
fpr_of	0.09	0.0100	53.0M
fpr_neg	0.00	0.0000	7.8M
fpr_sqr	0.00	0.0000	7.1M
fpr_sqr	0.00	0.0000	6.5M

Other — 1.8% of total

Function	%	self(s)	calls
poly_big_to_fp	0.60	0.0700	591.2K
modp_mkgm2	0.17	0.0200	996.0K
poly_sub_scaled	0.17	0.0200	521.1K
make_fg_step.constprop.0	0.17	0.0200	36.3K
modp_ninv31	0.09	0.0100	2.8M
poly_sub_scaled_ntt	0.09	0.0100	56.0K
poly_small_sqnorm	0.09	0.0100	21.3K
PQCLEAN_FALCON512_CLEAN_poly_invnorm2_fft	0.09	0.0100	13.4K
make_fg	0.09	0.0100	8.0K
make_fg_step	0.09	0.0100	8.0K
solve_NTRU_intermediate	0.09	0.0100	7.0K
poly_big_to_fp.constprop.1	0.09	0.0100	2.0K
mq_montysqr	0.00	0.0000	5.8M
modp_R	0.00	0.0000	5.8M
fpr_inv	0.00	0.0000	1.9M

4.2 Falcon-512 — Sign

-00 build (total: 15.930 s)

Hash / PRNG — 1.4% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_prng_refill	1.19	0.1900	39.1K
prng_get_u64	0.13	0.0200	1.8M
PQCLEAN_FALCON512_CLEAN_hash_to_point_ct	0.13	0.0200	1.0K
prng_get_u8	0.00	0.0000	5.3M
keccak_inc_squeeze	0.00	0.0000	722.1K
shake256_inc_squeeze	0.00	0.0000	722.1K
KeccakF1600_StatePermute	0.00	0.0000	12.2K
get_rng_u64	0.00	0.0000	4.1K
keccak_inc_absorb	0.00	0.0000	3.0K
shake256_inc_absorb	0.00	0.0000	3.0K
PQCLEAN_randombytes	0.00	0.0000	2.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	2.0K
keccak_inc_finalize	0.00	0.0000	2.0K

Function	%	self(s)	calls
keccak_inc_init	0.00	0.0000	2.0K
shake256_inc_ctx_release	0.00	0.0000	2.0K

Integer Arithmetic — 2.1% of total

Function	%	self(s)	calls
mq_montymul	0.94	0.1500	21.0M
PQCLEAN_FALCON512_CLEAN_gaussian0_sampler	0.47	0.0700	1.8M
mq_add	0.38	0.0600	9.2M
mq_sub	0.06	0.0100	9.2M
PQCLEAN_FALCON512_CLEAN_sampler	0.06	0.0100	1.0M
modp_montymul	0.06	0.0100	975.9K
mq NTT	0.06	0.0100	3.0K
mq_iNTT	0.06	0.0100	1.0K
modp_add	0.00	0.0000	683.6K
modp_sub	0.00	0.0000	367.3K
zint_mod_small_unsigned	0.00	0.0000	49.8K
modp_set	0.00	0.0000	30.0K
zint_mod_small_signed	0.00	0.0000	28.2K
zint_add_mul_small	0.00	0.0000	21.6K
zint_norm_zero	0.00	0.0000	21.1K

Floating-Point / FFT — 96.2% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_fpr_add	50.85	8.1000	148.6M
FPR	15.94	2.5400	297.5M
PQCLEAN_FALCON512_CLEAN_fpr_mul	14.69	2.3400	134.0M
fpr_ulsh	2.70	0.4300	148.6M
PQCLEAN_FALCON512_CLEAN_fpr_scaled	2.32	0.3700	12.0M
fpr_ursh	2.07	0.3300	152.2M
PQCLEAN_FALCON512_CLEAN_fpr_div	1.69	0.2700	2.3M
PQCLEAN_FALCON512_CLEAN_fpr_expm_p63	1.32	0.2100	1.8M
PQCLEAN_FALCON512_CLEAN_fpr_sqrt	0.94	0.1500	512.0K
fpr_sub	0.91	0.1400	70.5M
PQCLEAN_FALCON512_CLEAN_FFT	0.44	0.0700	9.6K
PQCLEAN_FALCON512_CLEAN_poly_split_fft	0.38	0.0600	2.0M
PQCLEAN_FALCON512_CLEAN_poly_LDL_fft	0.38	0.0600	511.0K
fpr_trunc	0.31	0.0500	3.6M
fpr_half	0.25	0.0400	16.4M

Other — 0.3% of total

Function	%	self(s)	calls
BerExp	0.16	0.0300	1.8M
fpr_floor	0.06	0.0100	1.0M

Function	%	self(s)	calls
mq_montysqr	0.00	0.0000	5.6M
fpr_inv	0.00	0.0000	2.3M
mq_conv_small	0.00	0.0000	1.5M
fpr_irsh	0.00	0.0000	1.0M
mq_div_12289	0.00	0.0000	512.5K
mq_rshift1	0.00	0.0000	9.0K
smallints_to_fpr	0.00	0.0000	8.0K
modp_R	0.00	0.0000	6.7K
fpr_lt	0.00	0.0000	6.5K
PQCLEAN_FALCON512_CLEAN_trim_i8_decode	0.00	0.0000	3.0K
modp_ninv31	0.00	0.0000	2.8K
modp_norm	0.00	0.0000	2.0K
PQCLEAN_FALCON512_CLEAN_comp_encode	0.00	0.0000	1.0K

-03 build (total: 3.240 s)

Hash / PRNG — 2.8% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_prng_refill	1.23	0.0400	39.2K
prng_get_u8	0.62	0.0200	5.3M
prng_get_u64	0.62	0.0200	1.8M
KeccakF1600_StatePermute	0.31	0.0100	15.4K
keccak_inc_finalize	0.00	0.0000	777.4K
shake256_inc_squeeze	0.00	0.0000	775.4K
get_rng_u64	0.00	0.0000	57.4K
keccak_inc_absorb	0.00	0.0000	3.0K
shake256_inc_absorb	0.00	0.0000	3.0K
PQCLEAN_randombytes	0.00	0.0000	2.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	2.0K
keccak_inc_init	0.00	0.0000	2.0K
shake256_inc_ctx_release	0.00	0.0000	2.0K
shake256_inc_finalize	0.00	0.0000	2.0K
shake256_inc_init	0.00	0.0000	2.0K

Integer Arithmetic — 2.8% of total

Function	%	self(s)	calls
mq_montymul	1.23	0.0400	20.0M
PQCLEAN_FALCON512_CLEAN_gaussian0_sampler	0.93	0.0300	1.8M
modp_add	0.31	0.0100	682.5K
mq NTT	0.31	0.0100	3.0K
mq_add	0.00	0.0000	9.2M
mq_sub	0.00	0.0000	9.2M
mq_montymul.constprop.0	0.00	0.0000	1.0M
PQCLEAN_FALCON512_CLEAN_sampler	0.00	0.0000	1.0M
modp_montymul	0.00	0.0000	963.3K

Function	%	self(s)	calls
modp_sub	0.00	0.0000	365.7K
zint_mod_small_unsigned	0.00	0.0000	49.3K
zint_rebuild_CRT.constprop.2	0.00	0.0000	36.1K
zint_mod_small_signed	0.00	0.0000	27.7K
zint_add_mul_small	0.00	0.0000	21.6K
poly_big_to_small	0.00	0.0000	20.0K

Floating-Point / FFT — 94.4% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512.CLEAN_fpr_add	50.93	1.6500	148.7M
PQCLEAN_FALCON512.CLEAN_fpr_mul	16.05	0.5200	134.1M
FPR	5.56	0.1800	297.7M
PQCLEAN_FALCON512.CLEAN_fpr_div	4.32	0.1400	2.3M
fpr_ursh	4.01	0.1300	152.3M
fpr_trunc	3.40	0.1100	152.3M
PQCLEAN_FALCON512.CLEAN_fpr_scaled	2.78	0.0900	12.0M
PQCLEAN_FALCON512.CLEAN_fpr_expm_p63	1.54	0.0500	1.8M
PQCLEAN_FALCON512.CLEAN_poly_split_fft	0.93	0.0300	2.0M
PQCLEAN_FALCON512.CLEAN_fpr_sqrt	0.93	0.0300	512.0K
fpr_sub	0.62	0.0200	70.5M
fpr_half	0.62	0.0200	16.4M
PQCLEAN_FALCON512.CLEAN_poly_merge_fft	0.62	0.0200	1.0M
PQCLEAN_FALCON512.CLEAN_FFT	0.62	0.0200	9.6K
fpr_neg	0.31	0.0100	14.1M

Other — 0.0% of total

Function	%	self(s)	calls
mq_montysqr	0.00	0.0000	5.6M
fpr_inv	0.00	0.0000	2.3M
BerExp	0.00	0.0000	1.8M
mq_conv_small	0.00	0.0000	1.5M
fpr_floor	0.00	0.0000	1.0M
fpr_irsh	0.00	0.0000	1.0M
mq_div_12289	0.00	0.0000	512.5K
mq_rshift1	0.00	0.0000	9.0K
smallints_to_fpr	0.00	0.0000	8.0K
modp_R	0.00	0.0000	5.7K
PQCLEAN_FALCON512.CLEAN_trim_i8_decode	0.00	0.0000	3.0K
modp_ninv31	0.00	0.0000	2.8K
solve_NTRU_binary_depth1	0.00	0.0000	1.5K
PQCLEAN_FALCON512.CLEAN_comp_encode	0.00	0.0000	1.0K
PQCLEAN_FALCON512.CLEAN_complete_private	0.00	0.0000	1.0K

4.3 Falcon-512 — Verify

-00 build (total: 0.190 s)

Hash / PRNG — 0.0% of total

Function	%	self(s)	calls
keccak_inc_squeeze	0.00	0.0000	758.7K
shake256_inc_squeeze	0.00	0.0000	758.7K
get_rng_u64	0.00	0.0000	41.0K
KeccakF1600_StatePermute	0.00	0.0000	13.4K
prng_get_u8	0.00	0.0000	5.2K
keccak_inc_absorb	0.00	0.0000	2.0K
shake256_inc_absorb	0.00	0.0000	2.0K
prng_get_u64	0.00	0.0000	1.7K
keccak_inc_finalize	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_hash_to_point_ct	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_prng_refill	0.00	0.0000	38

Integer Arithmetic — 78.9% of total

Function	%	self(s)	calls
mq_montymul	42.11	0.0800	8.5M
mq_sub	15.79	0.0300	7.4M
mq_add	10.53	0.0200	6.9M
modp_add	5.26	0.0100	683.6K
mq_iNTT	5.26	0.0100	1.0K
modp_montymul	0.00	0.0000	975.9K
modp_sub	0.00	0.0000	367.3K
zint_mod_small_unsigned	0.00	0.0000	49.8K
modp_set	0.00	0.0000	30.0K
zint_mod_small_signed	0.00	0.0000	28.2K
zint_add_mul_small	0.00	0.0000	21.6K
zint_norm_zero	0.00	0.0000	21.1K
zint_sub	0.00	0.0000	21.1K
zint_add_scaled_mul_small	0.00	0.0000	18.1K
mkgauss	0.00	0.0000	10.3K

Floating-Point / FFT — 21.1% of total

Function	%	self(s)	calls
FPR	15.79	0.0300	927.0K
PQCLEAN_FALCON512_CLEAN_fpr_add	5.26	0.0100	432.9K
fpr_ursh	0.00	0.0000	436.3K

Function	%	self(s)	calls
fpr_ulsh	0.00	0.0000	432.9K
PQCLEAN_FALCON512_CLEAN_fpr_mul	0.00	0.0000	425.1K
fpr_sub	0.00	0.0000	183.4K
PQCLEAN_FALCON512_CLEAN_fpr_scaled	0.00	0.0000	64.4K
fpr_of	0.00	0.0000	52.5K
fpr_neg	0.00	0.0000	21.5K
fpr_half	0.00	0.0000	16.4K
fpr_sqr	0.00	0.0000	12.8K
fpr_of	0.00	0.0000	11.8K
fpr_sqr	0.00	0.0000	6.7K
fpr_sub	0.00	0.0000	6.2K
fpr_rint	0.00	0.0000	4.3K

Other — 0.0% of total

Function	%	self(s)	calls
mq_montysqr	0.00	0.0000	11.3K
mq_rshift1	0.00	0.0000	9.0K
modp_R	0.00	0.0000	6.7K
fpr_lt	0.00	0.0000	6.5K
fpr_inv	0.00	0.0000	4.1K
modp_ninv31	0.00	0.0000	2.8K
mq_conv_small	0.00	0.0000	2.6K
modp_norm	0.00	0.0000	2.0K
BerExp	0.00	0.0000	1.7K
fpr_floor	0.00	0.0000	1.0K
fpr_irsh	0.00	0.0000	1.0K
mq_div_12289	0.00	0.0000	1.0K
mq_poly_montymul_ntt	0.00	0.0000	1.0K
mq_poly_tomonty	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_comp_decode	0.00	0.0000	1.0K

-03 build (total: 0.050 s)

Hash / PRNG — 40.0% of total

Function	%	self(s)	calls
PQCLEAN_FALCON512_CLEAN_hash_to_point_ct	40.00	0.0200	1.0K
keccak_inc_finalize	0.00	0.0000	743.3K
shake256_inc_squeeze	0.00	0.0000	742.3K
get_rng_u64	0.00	0.0000	24.6K
KeccakF1600_StatePermute	0.00	0.0000	12.5K
prng_get_u8	0.00	0.0000	5.2K
keccak_inc_absorb	0.00	0.0000	2.0K
shake256_inc_absorb	0.00	0.0000	2.0K
prng_get_u64	0.00	0.0000	1.7K
keccak_inc_init	0.00	0.0000	1.0K

Function	%	self(s)	calls
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_prng_refill	0.00	0.0000	38
PQCLEAN_randombytes	0.00	0.0000	4

Integer Arithmetic — 60.0% of total

Function	%	self(s)	calls
mq_add	40.00	0.0200	6.9M
mq_montymul	20.00	0.0100	8.0M
mq_sub	0.00	0.0000	7.4M
modp_montymul	0.00	0.0000	963.3K
modp_add	0.00	0.0000	682.5K
mq_montymul.constprop.0	0.00	0.0000	513.5K
modp_sub	0.00	0.0000	365.7K
zint_mod_small_unsigned	0.00	0.0000	49.3K
zint_rebuild_CRT.constprop.2	0.00	0.0000	36.1K
zint_mod_small_signed	0.00	0.0000	27.7K
zint_add_mul_small	0.00	0.0000	21.6K
poly_big_to_small	0.00	0.0000	20.0K
zint_norm_zero	0.00	0.0000	20.0K
zint_add_scaled_mul_small	0.00	0.0000	18.1K
modp_set	0.00	0.0000	14.7K

Floating-Point / FFT — 0.0% of total

Function	%	self(s)	calls
FPR	0.00	0.0000	877.8K
fpr_trunc	0.00	0.0000	410.0K
fpr_ursh	0.00	0.0000	410.0K
PQCLEAN_FALCON512_CLEAN_fpr_add	0.00	0.0000	406.5K
PQCLEAN_FALCON512_CLEAN_fpr_mul	0.00	0.0000	403.6K
fpr_sub	0.00	0.0000	171.1K
PQCLEAN_FALCON512_CLEAN_fpr_scaled	0.00	0.0000	63.4K
fpr_of	0.00	0.0000	51.5K
fpr_neg	0.00	0.0000	20.5K
fpr_half	0.00	0.0000	16.4K
fpr_of	0.00	0.0000	11.9K
fpr_sqr	0.00	0.0000	11.8K
fpr_sub	0.00	0.0000	6.2K
fpr_sqr	0.00	0.0000	5.7K
fpr_rint	0.00	0.0000	4.3K

Other — 0.0% of total

Function	%	self(s)	calls
mq_montysqr	0.00	0.0000	11.3K
mq_rshift1	0.00	0.0000	9.0K
modp_R	0.00	0.0000	5.7K
fpr_inv	0.00	0.0000	3.8K
modp_ninv31	0.00	0.0000	2.8K
mq_conv_small	0.00	0.0000	2.6K
BerExp	0.00	0.0000	1.7K
solve_NTRU_binary_depth1	0.00	0.0000	1.5K
fpr_floor	0.00	0.0000	1.0K
fpr_irsh	0.00	0.0000	1.0K
mq_div_12289	0.00	0.0000	1.0K
mq_poly_montymul_ntt	0.00	0.0000	1.0K
mq_poly_tomonty	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_comp_decode	0.00	0.0000	1.0K
PQCLEAN_FALCON512_CLEAN_crypto_sign_verify	0.00	0.0000	1.0K

4.4 Falcon-1024 — KeyGen

-00 build (total: 143.260 s)

Hash / PRNG — 3.2% of total

Function	%	self(s)	calls
KeccakF1600_StatePermute	2.03	2.9100	4.2M
keccak_inc_squeeze	0.94	1.3500	70.9M
get_rng_u64	0.12	0.1700	70.9M
shake256_inc_squeeze	0.07	0.1000	70.9M
shake256_inc_finalize	0.00	0.0100	1.0K
PQCLEAN_randombytes	0.00	0.0000	1.0K
keccak_inc_absorb	0.00	0.0000	1.0K
keccak_inc_finalize	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	1.0K
shake256_inc_absorb	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K

Integer Arithmetic — 33.3% of total

Function	%	self(s)	calls
modp_montymul	12.24	17.5400	2.80B
modp_add	5.65	8.0900	2.17B
zint_add_scaled_mul_small	3.36	4.8200	74.9M
zint_mod_small_unsigned	2.75	3.9400	112.4M
mkgauss	1.20	1.7200	35.5M
modp_NTT2_ext	1.12	1.6000	3.6M
modp_sub	1.01	1.4500	892.4M

Function	%	self(s)	calls
zint_finish_mod	0.96	1.3800	1.9M
zint_add_mul_small	0.93	1.3300	49.2M
modp_R2	0.87	1.2500	7.4M
zint_co_reduce_mod	0.58	0.8300	940.0K
zint_norm_zero	0.39	0.5600	47.4M
zint_bezout	0.34	0.4800	1.1K
modp_iNTT2_ext	0.29	0.4200	3.4M
zint_sub	0.28	0.4000	47.4M

Floating-Point / FFT — 62.5% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_add	32.73	46.8900	875.3M
FPR	11.49	16.4600	1.84B
PQCLEAN_FALCON1024_CLEAN_fpr_mul	10.04	14.3800	835.4M
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	3.49	5.0000	124.6M
fpr_ulsh	1.79	2.5600	875.3M
fpr_ursh	1.51	2.1700	875.3M
PQCLEAN_FALCON1024_CLEAN_fpr_div	0.38	0.5400	5.5M
fpr_sub	0.30	0.4300	366.1M
PQCLEAN_FALCON1024_CLEAN_FFT	0.30	0.4300	1.2M
PQCLEAN_FALCON1024_CLEAN_iFFT	0.30	0.4300	595.8K
fpr_rint	0.07	0.1000	9.1M
fpr_neg	0.04	0.0600	23.0M
PQCLEAN_FALCON1024_CLEAN_poly_mul_fft	0.02	0.0300	1.2M
fpr_of	0.01	0.0200	124.6M
fpr_ulsh	0.01	0.0200	9.1M

Other — 1.0% of total

Function	%	self(s)	calls
poly_big_to_fp	0.22	0.3200	1.2M
poly_sub_scaled	0.13	0.1900	1.1M
modp_div	0.10	0.1500	1.8M
modp_mkgm2	0.10	0.1400	1.8M
make_fg_step	0.08	0.1200	56.5K
modp_ninv31	0.07	0.1000	5.5M
solve_NTRU_intermediate	0.07	0.1000	8.2K
poly_sub_scaled_ntt	0.06	0.0800	56.1K
fpr_lt	0.05	0.0700	14.1M
mq_div_12289	0.02	0.0300	1.2M
PQCLEAN_FALCON1024_CLEAN_poly_mul_autoadjoint	0.02	0.0300	592.8K
poly_small_to_fp	0.02	0.0300	17.6K
PQCLEAN_FALCON1024_CLEAN_keygen	0.02	0.0300	1.0K
modp_Rx	0.01	0.0200	1.8M
poly_small_sqnorm	0.01	0.0100	34.6K

-03 build (total: 34.310 s)

Hash / PRNG — 4.5% of total

Function	%	self(s)	calls
KeccakF1600_StatePermute	2.74	0.9400	4.5M
get_rng_u64	1.02	0.3500	76.1M
keccak_inc_finalize	0.73	0.2500	76.1M
shake256_inc_squeeze	0.03	0.0100	76.1M
shake256_inc_init	0.03	0.0100	1.0K
PQCLEAN_randombytes	0.00	0.0000	1.0K
keccak_inc_absorb	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	1.0K
shake256_inc_absorb	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K

Integer Arithmetic — 40.6% of total

Function	%	self(s)	calls
modp_montymul	20.23	6.9400	2.81B
zint_add_scaled_mul_small	3.12	1.0700	75.2M
modp_add	2.97	1.0200	2.20B
mkgauss	2.07	0.7100	38.1M
zint_add_mul_small	1.66	0.5700	49.6M
zint_rebuild_CRT.constprop.2	1.54	0.5300	76.3M
zint_finish_mod	1.43	0.4900	1.9M
zint_mod_small_unsigned	1.34	0.4600	112.2M
modp_sub	1.25	0.4300	894.4M
zint_co_reduce_mod	0.87	0.3000	959.0K
modp_iNTT2_ext	0.76	0.2600	3.5M
zint_rebuild_CRT.constprop.3	0.44	0.1500	203.7K
zint_co_reduce	0.41	0.1400	479.5K
zint_mod_small_signed	0.32	0.1100	62.6M
zint_mul_small	0.32	0.1100	3.1M

Floating-Point / FFT — 53.2% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_add	29.12	9.9900	908.5M
PQCLEAN_FALCON1024_CLEAN_fpr_mul	11.75	4.0300	862.8M
FPR	3.67	1.2600	1.90B
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	2.27	0.7800	126.3M
fpr_trunc	1.81	0.6200	908.5M
fpr_ursh	1.75	0.6000	908.5M
PQCLEAN_FALCON1024_CLEAN_fpr_div	1.05	0.3600	5.8M
PQCLEAN_FALCON1024_CLEAN_FFT	0.64	0.2200	1.2M

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_iFFT	0.44	0.1500	605.3K
fpr_sub	0.35	0.1200	381.4M
fpr_of	0.17	0.0600	126.3M
fpr_rint	0.17	0.0600	9.1M
fpr_ulsh	0.03	0.0100	9.1M
fpr_neg	0.00	0.0000	24.1M
fpr_sqr	0.00	0.0000	22.1M

Other — 1.6% of total

Function	%	self(s)	calls
poly_big_to_fp	0.35	0.1200	1.2M
solve_NTRU_intermediate	0.26	0.0900	8.3K
modp_mkgm2	0.23	0.0800	1.9M
poly_sub_scaled	0.23	0.0800	1.1M
modp_div	0.15	0.0500	1.9M
PQCLEAN_FALCON1024_CLEAN_keygen	0.06	0.0200	1.0K
modp_ninv31	0.03	0.0100	5.6M
solve_NTRU_binary_depth1	0.03	0.0100	3.1M
modp_Rx	0.03	0.0100	1.8M
poly_sub_scaled_ntt	0.03	0.0100	56.2K
make_fg_step.constprop.0	0.03	0.0100	47.9K
poly_small_sqnorm	0.03	0.0100	37.1K
poly_small_to_fp	0.03	0.0100	18.6K
poly_big_to_fp.constprop.1	0.03	0.0100	2.0K
solve_NTRU	0.03	0.0100	1.1K

4.5 Falcon-1024 — Sign

-00 build (total: 34.240 s)

Hash / PRNG — 1.8% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_prng_refill	1.26	0.4300	76.6K
PQCLEAN_FALCON1024_CLEAN_hash_to_point_ct0.23	0.23	0.0800	1.0K
prng_get_u64	0.13	0.0400	3.5M
KeccakF1600_StatePermute	0.09	0.0300	22.2K
prng_get_u8	0.04	0.0100	10.5M
keccak_inc_squeeze	0.03	0.0100	1.3M
shake256_inc_squeeze	0.00	0.0000	1.3M
get_rng_u64	0.00	0.0000	20.5K
keccak_inc_absorb	0.00	0.0000	3.0K
shake256_inc_absorb	0.00	0.0000	3.0K
PQCLEAN_randombytes	0.00	0.0000	2.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	2.0K
keccak_inc_finalize	0.00	0.0000	2.0K

Function	%	self(s)	calls
keccak_inc_init	0.00	0.0000	2.0K
shake256_inc_ctx_release	0.00	0.0000	2.0K

Integer Arithmetic — 2.0% of total

Function	%	self(s)	calls
mq_montymul	0.88	0.3000	44.1M
PQCLEAN_FALCON1024_CLEAN_gaussian0_sampler	0.38	0.1300	3.5M
mq_add	0.34	0.1200	20.5M
PQCLEAN_FALCON1024_CLEAN_sampler	0.18	0.0600	2.0M
mq NTT	0.09	0.0300	3.0K
modp_montymul	0.06	0.0200	2.7M
mq_sub	0.03	0.0100	20.5M
mq_iNTT	0.03	0.0100	1.0K
modp_add	0.00	0.0000	2.1M
modp_sub	0.00	0.0000	879.0K
zint_mod_small_unsigned	0.00	0.0000	110.0K
zint_add_scaled_mul_small	0.00	0.0000	74.5K
modp_set	0.00	0.0000	62.1K
zint_mod_small_signed	0.00	0.0000	61.9K
zint_add_mul_small	0.00	0.0000	48.1K

Floating-Point / FFT — 96.0% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_add	50.22	17.1900	330.0M
FPR	17.79	6.0900	653.4M
PQCLEAN_FALCON1024_CLEAN_fpr_mul	13.99	4.7900	293.3M
fpr_ulsh	2.66	0.9100	330.0M
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	2.54	0.8700	23.9M
fpr_ursh	1.87	0.6400	337.0M
PQCLEAN_FALCON1024_CLEAN_fpr_div	1.72	0.5900	5.1M
PQCLEAN_FALCON1024_CLEAN_fpr_expm_p63	1.17	0.4000	3.5M
PQCLEAN_FALCON1024_CLEAN_fpr_sqrt	0.91	0.3100	1.0M
PQCLEAN_FALCON1024_CLEAN_FFT	0.61	0.2100	10.1K
fpr_sub	0.51	0.1700	157.6M
PQCLEAN_FALCON1024_CLEAN_poly_split_fft	0.35	0.1200	4.1M
PQCLEAN_FALCON1024_CLEAN_poly_LDL_fft	0.26	0.0900	1.0M
fpr_half	0.22	0.0700	36.9M
fpr_trunc	0.20	0.0700	7.0M

Other — 0.3% of total

Function	%	self(s)	calls
BerExp	0.06	0.0200	3.5M
mq_div_12289	0.06	0.0200	1.0M

Function	%	self(s)	calls
smallints_to_fpr	0.06	0.0200	8.0K
mq_montysqr	0.03	0.0100	11.3M
PQCLEAN_FALCON1024_CLEAN_comp_encode	0.03	0.0100	1.0K
PQCLEAN_FALCON1024_CLEAN_complete_private	0.03	0.0100	1.0K
mq_conv_small	0.01	0.0100	3.1M
fpr_inv	0.00	0.0000	5.1M
fpr_floor	0.00	0.0000	2.0M
fpr_irsh	0.00	0.0000	2.0M
fpr_lt	0.00	0.0000	14.0K
modp_R	0.00	0.0000	12.0K
mq_rshift1	0.00	0.0000	10.0K
modp_ninv31	0.00	0.0000	5.2K
modp_norm	0.00	0.0000	4.1K

-03 build (total: 6.810 s)

Hash / PRNG — 1.2% of total

Function	%	self(s)	calls
prng_get_u64	0.29	0.0200	3.5M
PQCLEAN_FALCON1024_CLEAN_prng_refill	0.29	0.0200	76.6K
PQCLEAN_FALCON1024_CLEAN_hash_to_point_ct	0.29	0.0200	1.0K
prng_get_u8	0.15	0.0100	10.5M
keccak_inc_finalize	0.15	0.0100	1.3M
shake256_inc_squeeze	0.00	0.0000	1.3M
KeccakF1600_StatePermute	0.00	0.0000	21.7K
get_rng_u64	0.00	0.0000	12.3K
keccak_inc_absorb	0.00	0.0000	3.0K
shake256_inc_absorb	0.00	0.0000	3.0K
PQCLEAN_randombytes	0.00	0.0000	2.0K
randombytes_linux_randombytes_getrandom	0.00	0.0000	2.0K
keccak_inc_init	0.00	0.0000	2.0K
shake256_inc_ctx_release	0.00	0.0000	2.0K
shake256_inc_finalize	0.00	0.0000	2.0K

Integer Arithmetic — 3.2% of total

Function	%	self(s)	calls
mq_montymul	1.03	0.0700	42.0M
mq_NTT	0.59	0.0400	3.0K
PQCLEAN_FALCON1024_CLEAN_gaussian0_sampler	0.44	0.0300	3.5M
mq_add	0.29	0.0200	20.5M
PQCLEAN_FALCON1024_CLEAN_sampler	0.29	0.0200	2.0M
mq_sub	0.15	0.0100	20.5M
modp_montymul	0.15	0.0100	2.7M
zint_mod_small_unsigned	0.15	0.0100	108.9K
mq_iNTT	0.15	0.0100	1.0K

Function	%	self(s)	calls
modp_add	0.00	0.0000	2.1M
mq_montymul.constprop.0	0.00	0.0000	2.0M
modp_sub	0.00	0.0000	875.9K
zint_rebuild_CRT.constprop.2	0.00	0.0000	75.0K
zint_add_scaled_mul_small	0.00	0.0000	74.5K
zint_mod_small_signed	0.00	0.0000	60.9K

Floating-Point / FFT — 95.6% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_add	49.05	3.3400	329.9M
PQCLEAN_FALCON1024_CLEAN_fpr_mul	21.88	1.4900	293.3M
FPR	6.17	0.4200	653.2M
PQCLEAN_FALCON1024_CLEAN_fpr_div	4.55	0.3100	5.1M
fpr_ursh	3.08	0.2100	336.9M
fpr_trunc	1.76	0.1200	336.9M
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	1.62	0.1100	23.9M
PQCLEAN_FALCON1024_CLEAN_fpr_expm_p63	1.62	0.1100	3.5M
PQCLEAN_FALCON1024_CLEAN_fpr_sqrt	1.03	0.0700	1.0M
PQCLEAN_FALCON1024_CLEAN_poly_split_fft	0.88	0.0600	4.1M
fpr_sub	0.73	0.0500	157.6M
PQCLEAN_FALCON1024_CLEAN_poly_LDL_fft	0.73	0.0500	1.0M
PQCLEAN_FALCON1024_CLEAN_FFT	0.73	0.0500	10.1K
fpr_half	0.29	0.0200	36.9M
fpr_neg	0.29	0.0200	30.7M

Other — 0.0% of total

Function	%	self(s)	calls
mq_montysqr	0.00	0.0000	11.3M
fpr_inv	0.00	0.0000	5.1M
BerExp	0.00	0.0000	3.5M
mq_conv_small	0.00	0.0000	3.1M
fpr_floor	0.00	0.0000	2.0M
fpr_irsh	0.00	0.0000	2.0M
mq_div_12289	0.00	0.0000	1.0M
modp_R	0.00	0.0000	10.3K
mq_rshift1	0.00	0.0000	10.0K
smallints_to_fpr	0.00	0.0000	8.0K
modp_ninv31	0.00	0.0000	5.2K
solve_NTRU_binary_depth1	0.00	0.0000	3.1K
PQCLEAN_FALCON1024_CLEAN_trim_i8_decode	0.00	0.0000	3.0K
modp_div	0.00	0.0000	1.7K
modp_mkgm2	0.00	0.0000	1.7K

4.6 Falcon-1024 — Verify

-00 build (total: 0.590 s)

Hash / PRNG — 13.6% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_hash_to_point_ct1	1.86	0.0700	1.0K
KeccakF1600_StatePermute	1.69	0.0100	31.8K
keccak_inc_squeeze	0.00	0.0000	1.5M
shake256_inc_squeeze	0.00	0.0000	1.5M
get_rng_u64	0.00	0.0000	200.9K
prng_get_u8	0.00	0.0000	10.4K
prng_get_u64	0.00	0.0000	3.5K
keccak_inc_absorb	0.00	0.0000	2.0K
shake256_inc_absorb	0.00	0.0000	2.0K
keccak_inc_finalize	0.00	0.0000	1.0K
keccak_inc_init	0.00	0.0000	1.0K
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_prng_refill	0.00	0.0000	76

Integer Arithmetic — 54.2% of total

Function	%	self(s)	calls
mq_montymul	23.73	0.1400	18.5M
mq_add	8.47	0.0500	15.4M
modp_montymul	5.08	0.0300	3.9M
modp_add	3.39	0.0200	3.1M
mq NTT	3.39	0.0200	2.0K
mq_sub	1.69	0.0100	16.4M
zint_mod_small_unsigned	1.69	0.0100	135.8K
zint_add_scaled_mul_small	1.69	0.0100	76.9K
zint_add_mul_small	1.69	0.0100	60.2K
mq_iNTT	1.69	0.0100	1.0K
poly_small_mkgauss	1.69	0.0100	98
modp_sub	0.00	0.0000	1.0M
mkgauss	0.00	0.0000	100.4K
zint_mod_small_signed	0.00	0.0000	75.6K
modp_set	0.00	0.0000	68.2K

Floating-Point / FFT — 28.8% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_add	18.64	0.1100	2.2M
FPR	5.08	0.0300	4.3M
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	1.69	0.0100	191.6K

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_poly_LDL_fft	1.69	0.0100	1.0K
PQCLEAN_FALCON1024_CLEAN_iFFT	1.69	0.0100	875
fpr_ursh	0.00	0.0000	2.2M
fpr_ulsh	0.00	0.0000	2.2M
PQCLEAN_FALCON1024_CLEAN_fpr_mul	0.00	0.0000	1.9M
fpr_sub	0.00	0.0000	972.5K
fpr_of	0.00	0.0000	167.9K
fpr_neg	0.00	0.0000	86.9K
fpr_sqr	0.00	0.0000	67.6K
fpr_sqr	0.00	0.0000	55.4K
fpr_half	0.00	0.0000	36.9K
fpr_of	0.00	0.0000	23.7K

Other — 3.4% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_comp_decode	3.39	0.0200	1.0K
mq_montysqr	0.00	0.0000	33.8K
modp_R	0.00	0.0000	20.7K
fpr_inv	0.00	0.0000	18.9K
fpr_lt	0.00	0.0000	15.1K
mq_rshift1	0.00	0.0000	10.0K
modp_ninv31	0.00	0.0000	9.0K
mq_conv_small	0.00	0.0000	7.2K
modp_norm	0.00	0.0000	4.1K
BerExp	0.00	0.0000	3.5K
mq_div_12289	0.00	0.0000	3.1K
modp_div	0.00	0.0000	2.9K
modp_mkgm2	0.00	0.0000	2.9K
modp_Rx	0.00	0.0000	2.8K
fpr_floor	0.00	0.0000	2.0K

-03 build (total: 0.080 s)

Hash / PRNG — 25.0% of total

Function	%	self(s)	calls
KeccakF1600_StatePermute	12.50	0.0100	25.3K
PQCLEAN_FALCON1024_CLEAN_hash_to_point_ct	12.50	0.0100	1.0K
keccak_inc_finalize	0.00	0.0000	1.4M
shake256_inc_squeeze	0.00	0.0000	1.4M
get_rng_u64	0.00	0.0000	90.2K
prng_get_u8	0.00	0.0000	10.7K
prng_get_u64	0.00	0.0000	3.6K
keccak_inc_absorb	0.00	0.0000	2.0K
shake256_inc_absorb	0.00	0.0000	2.0K
keccak_inc_init	0.00	0.0000	1.0K

Function	%	self(s)	calls
shake256_inc_ctx_release	0.00	0.0000	1.0K
shake256_inc_finalize	0.00	0.0000	1.0K
shake256_inc_init	0.00	0.0000	1.0K
PQCLEAN_FALCON1024_CLEAN_prng_refill	0.00	0.0000	78
PQCLEAN_randombytes	0.00	0.0000	4

Integer Arithmetic — 62.5% of total

Function	%	self(s)	calls
mq_add	25.00	0.0200	15.4M
mq_montymul	12.50	0.0100	17.5M
mq NTT	12.50	0.0100	2.0K
mq_iNTT	12.50	0.0100	1.0K
mq_sub	0.00	0.0000	16.4M
modp_montymul	0.00	0.0000	3.9M
modp_add	0.00	0.0000	3.1M
mq_montymul.constprop.0	0.00	0.0000	1.0M
modp_sub	0.00	0.0000	1.0M
zint_mod_small_unsigned	0.00	0.0000	134.8K
zint_rebuild_CRT.constprop.2	0.00	0.0000	84.6K
zint_add_scaled_mul_small	0.00	0.0000	77.0K
zint_mod_small_signed	0.00	0.0000	74.6K
zint_add_mul_small	0.00	0.0000	60.2K
poly_big_to_small	0.00	0.0000	56.4K

Floating-Point / FFT — 12.5% of total

Function	%	self(s)	calls
PQCLEAN_FALCON1024_CLEAN_fpr_mul	12.50	0.0100	1.2M
FPR	0.00	0.0000	2.7M
fpr_trunc	0.00	0.0000	1.3M
fpr_ursh	0.00	0.0000	1.3M
PQCLEAN_FALCON1024_CLEAN_fpr_add	0.00	0.0000	1.3M
fpr_sub	0.00	0.0000	557.8K
PQCLEAN_FALCON1024_CLEAN_fpr_scaled	0.00	0.0000	161.4K
fpr_of	0.00	0.0000	137.4K
fpr_neg	0.00	0.0000	56.2K
fpr_sqr	0.00	0.0000	36.9K
fpr_half	0.00	0.0000	36.9K
fpr_sqr	0.00	0.0000	24.6K
fpr_of	0.00	0.0000	24.0K
fpr_sub	0.00	0.0000	12.8K
PQCLEAN_FALCON1024_CLEAN_fpr_div	0.00	0.0000	11.3K

Other — 0.0% of total

Function	%	self(s)	calls
mq_montysqr	0.00	0.0000	33.8K
modp_R	0.00	0.0000	17.7K
fpr_inv	0.00	0.0000	11.3K
mq_rshift1	0.00	0.0000	10.0K
modp_ninv31	0.00	0.0000	9.0K
mq_conv_small	0.00	0.0000	7.2K
BerExp	0.00	0.0000	3.6K
solve_NTRU_binary_depth1	0.00	0.0000	3.1K
mq_div_12289	0.00	0.0000	3.1K
modp_div	0.00	0.0000	2.9K
modp_mkgm2	0.00	0.0000	2.9K
modp_Rx	0.00	0.0000	2.8K
fpr_floor	0.00	0.0000	2.0K
fpr_irsh	0.00	0.0000	2.0K
poly_big_to_fp	0.00	0.0000	1.7K

5 Key Findings

- **KeyGen is dominated by Integer NTT and FP/FFT arithmetic.** `modp_montymul` (Montgomery multiplication for NTT) and `fpr_add` (FFT floating-point add) together account for over 40% of total keygen time. The lattice basis generation (`zint_*`, `mkgauss`) contributes a further ~10–15%.
- **Sign is almost entirely FP/FFT.** The `ffSampling_fft_dyntree` recursive sampler drives repeated calls to `fpr_add` (~50%) and `fpr_mul` (~15%). Integer NTT (`mq_montymul`) is visible but minor (<2%).
- **Verify is entirely Hash and Integer NTT.** No floating-point FFT is used. Time splits between `hash_to_point_ct` (SHAKE-256 hashing of the message) and `mq_montymul` / `mq_NTT` (NTT-based polynomial multiplication over $\mathbb{Z}_q$). This is why verify is 70–300× faster than keygen.
- **Falcon-1024 costs $\approx 2\text{--}3\times$ more than Falcon-512.** The FFT and NTT algorithms are $\mathcal{O}(N \log N)$; doubling N from 512 to 1024 roughly doubles the cost with an additional log factor.
- **O3 gives $4\text{--}5\times$ speedup on keygen/sign, $7\times$ on verify.** The FPR constructor overhead drops from ~15% to <5% under O3 (compiler promotes the struct boxing). Verify benefits most from O3 because `mq_montymul` vectorises well.